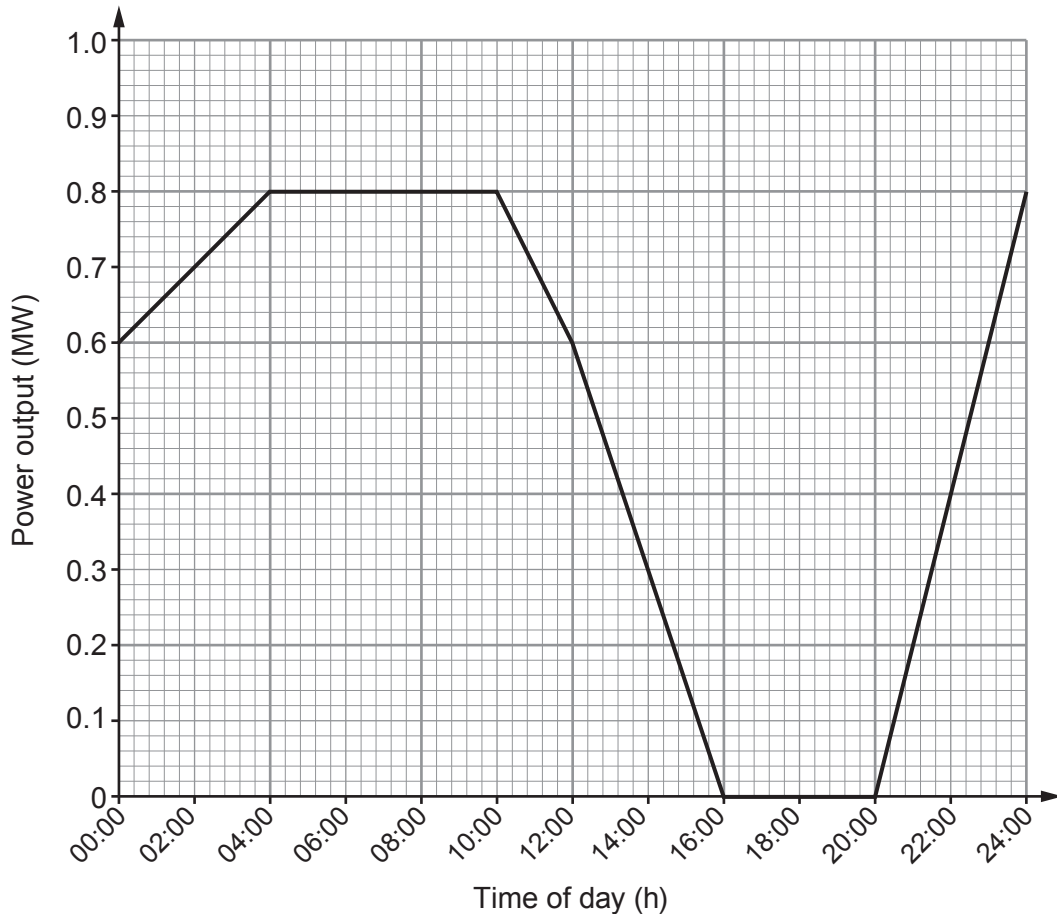


3. One renewable energy source that large energy suppliers use to generate electricity is wind. The percentage of electricity generated in this way has been gradually increasing over the last few years.

The graph below shows the power output of a wind turbine in one day. This wind turbine has a mean output of 0.6 MW.



- (a) Use the graph and the equation:

$$\text{units generated (kWh)} = \text{power (kW)} \times \text{time (h)}$$

to calculate the number of units generated between 04:00 and 10:00.

[4]

(1 MW = 1000 kW)

number of units = kWh



- (b) The table below gives information about generating electricity by wind and a nuclear power station.

	wind turbine	nuclear power station
expected lifetime (years)	20	60
mean power output (MW)	0.60	3 600
land area needed (km ²)	0.55	4.5
cost to commission (£ million)	3	4 000
waste produced	none	radioactive waste
lifetime carbon footprint (gCO ₂ /kWh)	5	5
overall cost of generating electricity (p/kWh)	5.6	2.8

Use the table to answer the following questions.

- (i) Calculate the number of wind turbines that need to be built to provide the same power as the nuclear power station will provide in its lifetime. [2]

number of wind turbines =

- (ii) Calculate the land area needed by a wind farm to produce the same power as one nuclear power station. [3]

land area = km²



- (c) (i) Evaluate the cost effectiveness of the two methods of generating electricity. [3]

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- (ii) Evaluate the environmental impact of the two methods of generating electricity. [2]

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- (d) The lifetime carbon footprints of a wind farm and a nuclear power station are 5gCO₂/kWh.

Suggest how the carbon footprint of a coal-fired power station might compare to this value.

Give a reason for your answer.

[2]

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